DIGI Patient Interface Module

PHILIPS

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Catheter

J6 — 53047

1	CATHSTR
2	EEPROM
3	PIM+
4	PIM-
5	PIM_GND
6	PIM_HV
7	OUT+
8	OUT-
9	TR+
10	TR-
11	CLK
12	CATH+10V

CLK

ASIC_Interface

PIM+, PIM-, PIM_GND, PIM_HV, OUT+, OUT-, TR+, TR-, CLK, CATH+10V

TR_SWI_CONTROL, SCK, /DACS, SDAT, DGCTRIG, DGCTRIG2, TGCA0, TGCA1, RXAMPSW

XF7, XF8

LGCHV_RAIL_EN, EN_VMAG

RF+, RF-, TestIN+, TestIN-

FGA_PD3, FGA_PD1_N, FGA_PD2_N

+HV, 9.0VDC, 7.0VDC, P/N, N/P

DAT+, DAT-, DIR_R/L, XMIT, /CLOCK, NOPROBE, LGCHV_OI_ACTL, STAT_HI, STAT_LO, XF9

SHT 2-5

TestIN+, TestIN-

Analog_Processing

RF+, RF-

ADCP0, ADCN0, ADCP1, ADCN1, ADCP2, ADCN2, ADCP3, ADCN3, ADCP4, ADCN4, ADCP5, ADCN5, ADCP6, ADCN6

CLK+_ADC, CLK-_ADC, OF_+, OF_-, /CS_ADC, SCK_ADC, SDI_ADC, SDO_ADC

100MHz_SYS_CLK+, 100MHz_SYS_CLK-

P/N, N/P, DAT+, DAT-, DIR_R/L, XMIT, /CLOCK, NOPROBE, LGCHV_OI_ACTL, STAT_LO, STAT_HI, HVADJ_DIN, HVADJ_SCLK, HVADJ_SYNC, HV_OVFAULT_DIN, HV_OVFAULT_SCLK, HV_OVFAULT_SYNC

TR_SWI_CONTROL, SCK, /DACS, SDAT, DGCTRIG2, DGCTRIG, TGCA0, TGCA1, RXAMPSW, LGCHV_RAIL_EN, HVDACCSI, HVDACSCI, HVDACSDI, HVCLIF, EN_VMAG, PRDHV_RAIL_EN

SHT 9

Digital_Processing

TestIN+, TestIN-, ADCP0, ADCN0, ADCP1, ADCN1, ADCP2, ADCN2, ADCP3, ADCN3, ADCP4, ADCN4, ADCP5, ADCN5, ADCP6, ADCN6, CLK+_ADC, CLK-_ADC, OF_+, OF_-, /CS_ADC, SCK_ADC, SDI_ADC, SDO_ADC, 100MHz_SYS_CLK+, 100MHz_SYS_CLK-, P/N, N/P, DAT+, DAT-, DIR_R/L, XMIT, /CLOCK, NOPROBE, LGCHV_OI_ACTL, STAT_LO, STAT_HI, HVADJ_DIN, HVADJ_SCLK, HVADJ_SYNC, HV_OVFAULT_DIN, HV_OVFAULT_SCLK, HV_OVFAULT_SYNC, TR_SWI_CONTROL, SCK, /DACS, SDAT, DGCTRIG2, DGCTRIG, TGCA0, TGCA1, RXAMPSW, LGCHV_RAIL_EN, HVDACCSI, HVDACSCI, HVDACSDI, HVCLIF, EN_VMAG, PRDHV_RAIL_EN, PRDHV_OI_ACTL, PRDHV_CRBR_EN, XF3, PRDHV_UV_ACTL, PRDHV_OV_ACTL, FRES, XF7, XF8, XF9, +12V_PROD_EN, +12V_ATREN, CATH3P3V_OI_ACTL, CATH3P3V_RAIL_EN, 3V3MON, HVDADJ_DIN, HVDADJ_SCLK, HVDADJ_SYNC

SHT 10-14

Ethernet&Power_Isolation

PHY_TX_CLK, PHY_TX_CTRL, PHY_TXD0, PHY_TXD1, PHY_TXD2, PHY_TXD3, PHY_RX_CLK, PHY_RX_CTRL, PHY_RXD0, PHY_RXD1, PHY_RXD2, PHY_RXD3, PHY_MDC, PHY_MDIO, PHY_RESETn, PHY_DVDD, +3.3V_PHY, +1.8V_PHY, +5V_ISO

SHT 15

1Gbit Ethernet

J5 — HFJT1

1	DA+
2	DA-
3	DB+
4	DB-
5	DC+
6	DC-
7	DD+
8	DD-
9	SHIELD

K1 (GREEN), K2 (YELLOW), A1, A2, I504, FB1 600R WE-CBF, GND

DO NOT MOUNT X5 — WR-PHD

J7 — 2-1445055-4

5V_SYSIN, GND

Power Input

CLK, CATHSTR, EEPROM, FGA_PD1_N, FGA_PD2_N, FGA_PD3, SYNC_300

Local_Power

+3.3Vcco, +2.5Vin, +1.8Vin, +1.2Vin, +1.0Vin, +3.3VinA, +1.8VADC, FRES

SHT 16

ASIC_Supply

+3.3VinA, 5V_EN, 12V_EN

+12V, +3.3Vin, +5Vin, +5VA, +12VPIM, +12VDAT, HVCLIF

HVDACCSI, HVDACSCI, HVDACSDI, PRDHV_RAIL_EN, HVADJ_DIN, HVADJ_SCLK, HVADJ_SYNC, HV_OVFAULT_DIN, HV_OVFAULT_SCLK, HV_OVFAULT_SYNC, HV_UVFAULT_DIN, HV_UVFAULT_SCLK, HV_UVFAULT_SYNC, CATH3P3V_RAIL_EN

PRDHV_OI_ACTL, PRDHV_CRBR_EN, XF3, PRDHV_UV_ACTL, PRDHV_OV_ACTL

+12V_PROD_EN, +12V_ATREN, CATH3P3V_OI_ACTL, 3V3MON

SHT 6-8

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 3000 098 20164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A TOP BLOCK	3000 090 7728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * 2022-08-01
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 1/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 © 2025 ROYAL PHILIPS N.V.

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Switches shown deactivated (control input = 0)

*FW inverts signal

ATR Chip Control Protocol

OUT+	OUT-	OUT-	OUT-	FUNCTION
L	L			Transmit Load
H	L			Transmit Shift
L	H			Receive Load
H-	H+			Receive Shift Left
H+	H-			Receive Shift Right
L	L			Chip Reset

Notes:
L = 0V, H- = 7V, H+ = 9V
Receive Shift Left is the Forward Direction

*Power dissipation= 5mW @ 5mA
*17mV @ 150mA dropout Voltage

U3 MIC5207 — U9 MIC5207 — U5 ADG1434

*Low: Active
*High: Disabled

Per:1msec
Pulse Wid: 500usec
3.3V / 0VDC

DIR_R/L — RC0603 R22 100R 1% — RC0603 R14 100R 1% — XF9

D22 BAV99DW — BAV99DW D22

R23 249R 1% — OUT-
R15 249R 1% — OUT+

I10 8.88V 6.88V
I11 8.88V 6.88V

OUT+ OUT+ OUT+
OUT- OUT- OUT-

*FW inverts signal
Per:37usec Pulse
Wid: 1usec
3.3V
/CLOCK — /CLOCK

Q9 FDV301N

ESD diode to be placed near to J6 connector

D2 SRDA12

12V 500W

1usec 6.5Vpk
4V
CLK — CLK

DAT+ — DAT+
DAT- — DAT-

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A Catheter_control_1of2	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * 2022-08-01
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 2/16 — A3
REVIEWED BY XIAOYANG CHEN		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 — © 2025 ROYAL PHILIPS N.V.

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PHILIPS

5.0V
XMIT
XMIT
RC0402 R33 499R 1%
C52 68p 50V
R34 0R
I13
130R 1% R35
GND
C177 100n 50V C0603
T1 TC1-1TX+
0.43Vpk
C46 10n 50V GRM155R7
C53 10n 50V GRM155R7
0.43Vpk
GND GND
TP32
TP33
5Vpk 4.4V 3.8Vpk
TR+ 140uA TR+
TR- 140uA TR-
5Vpk 4.6V 3.8Vpk
TPD3E001 U17
IO3 IO2 IO1 VOLT GND VCC
I80
GND C119 100n 16V CC0402-X7R
+5VA
ESD diode to be placed near to J6 connector
TPG1
GND
+10Viso
C10 100n 50V C0603
20K 1% R3
67VDC
C21 100n 50V C0603
6R04 1% R26
RC0603 100R 1% R21
RC0603 100R 1% R7
Q4 MMBT3904
Q3 MMBT3904
RC0603 2K 1% R8
+3.3Vin
RC0603 10K 1% R1
C5 100n 50V C0603
NOPROBE
NOPROBE
High:3.3VDC No Cath
Low:0VDC Catheter
RC0603 R4 2K 1%
0.83VDC
RC0603 R2 2K 1%
Q1 MMBT3904
GND

+3.3Vin
C7 10u 16V
C18 1u 25V GRM155R6
TP68
GND GND
U6 LMP8646
IN+ IN- V+ V- VOUT RG
*Window Comparator for CATH+10V
GAIN = x10
1.50V = 15mA Output
1.25V = 12mA Output
1.00V = 10mA Output
0.05V = 5mA Output
Ref Volt are 1.25V to 0V.
*5msec delay - RC Circuit
RC0603 R19 49R9 1%
C226 100p 50V CC0402-NPO
64K9 1% R168
RC0603 10K 1% R13
C41 100n 16V CC0402-X7R
I15
R330 130R 1%
C23 100u 6.3V
+3.3Vin
C4 10u 16V
C1 10n 50V CC0603-X7R
U2 TLV3012 V+ V- REF
I16
RC0402 R338 100R 1%
CATHFAIL
LGCHV_OI_ACTL
*Using PIMSA Cable and Cath
expect inrush 1.60V 16mA for 2msec
GND GND GND GND GND GND GND

LGCHV_RAIL_EN LGCHV_RAIL_EN
+10VCATH

+12VDAT
I14
U7 MIC2951
IN OUT FB SNS SHDN TAP GND ERR
C9 10u 16V
C16 100n 50V C0603
RCG0603 R255 1K 1%
LGCHV_RAIL_EN LGCHV_RAIL_EN
High: Regulator OFF
Low: Regulator ON
+10Viso
R329 RC0603 0R
R20 RC0603 10R 1%
150K 1% R169
C44 100n 50V C0603
C231 10u 16V
C15 1u 25V GRM155R6
C20 100n 16V CC0402-X7R
C19 1n 50V 04025C
TP1 5015
RC0805 R374 0R DO NOT MOUNT
L47 600R HZ0805
C175 100p 50V CC0402-NPO
CATH+10V
Current: 100mA
GND GND GND GND GND GND GND
1.23VDC
RC0603 R170 0R R171 20K 1%
GND
NOTE: Select a value around the nominal 1.10k to trim the voltage if needed.
For low voltage ATR-X mode, change resistor values for 3.3V

GND
TPG3
GND

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A Catheter_control_2of2	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 * . * .
See cPLM See cPLM			
See cPLM See cPLM			
See cPLM See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 3/16 A3
REVIEWED BY XIAOYANG CHEN		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 © 2025 ROYAL PHILIPS N.V.

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*For Transit pulse: NP goes High 1st

PN
NP
EN

*For command pulse bit "1" PN then NP High then idle.

(EN not used any more)

Note: MD1213 has inverting outputs
Keep OE high, otherwise OUTA=H and OUTB=L

C93 for DC blocking

3V3 level compatible

Hi= RX, Lo=TX

Ton=80ns Toff =41ns

ESD diode to be placed near to J6 connector

DO NOT MOUNT

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A RF_frontend_1of2	30000907728	4 2025-07-07 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 4/16 A3
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PHILIPS

ATR*/Prodigy

2nd order Butterworth, BPF

3rd order Elliptic
High pass Filter, Fc=2 MHz
Fstop=1.6 MHz, 0.5dB ripple

gap of 200um between Wire track & L77

track should be at least 10mm long

3V3->5V level converter

Hi= Prodigy, Lo = EE
4 Ohm; 371MHz BW
C=14.5 pF

VMAG to compensate
ADC impedance matching

RF TX/RX for ATR* Catheter

VBDS	GAIN
1.5VDC	50dB
1.2VDC	40dB
1.0VDC	34dB
0.9VDC	30dB
0.8VDC	28dB
0.6VDC	23dB
0.2VDC	8dB

DGCTRIG Time Constant controlled

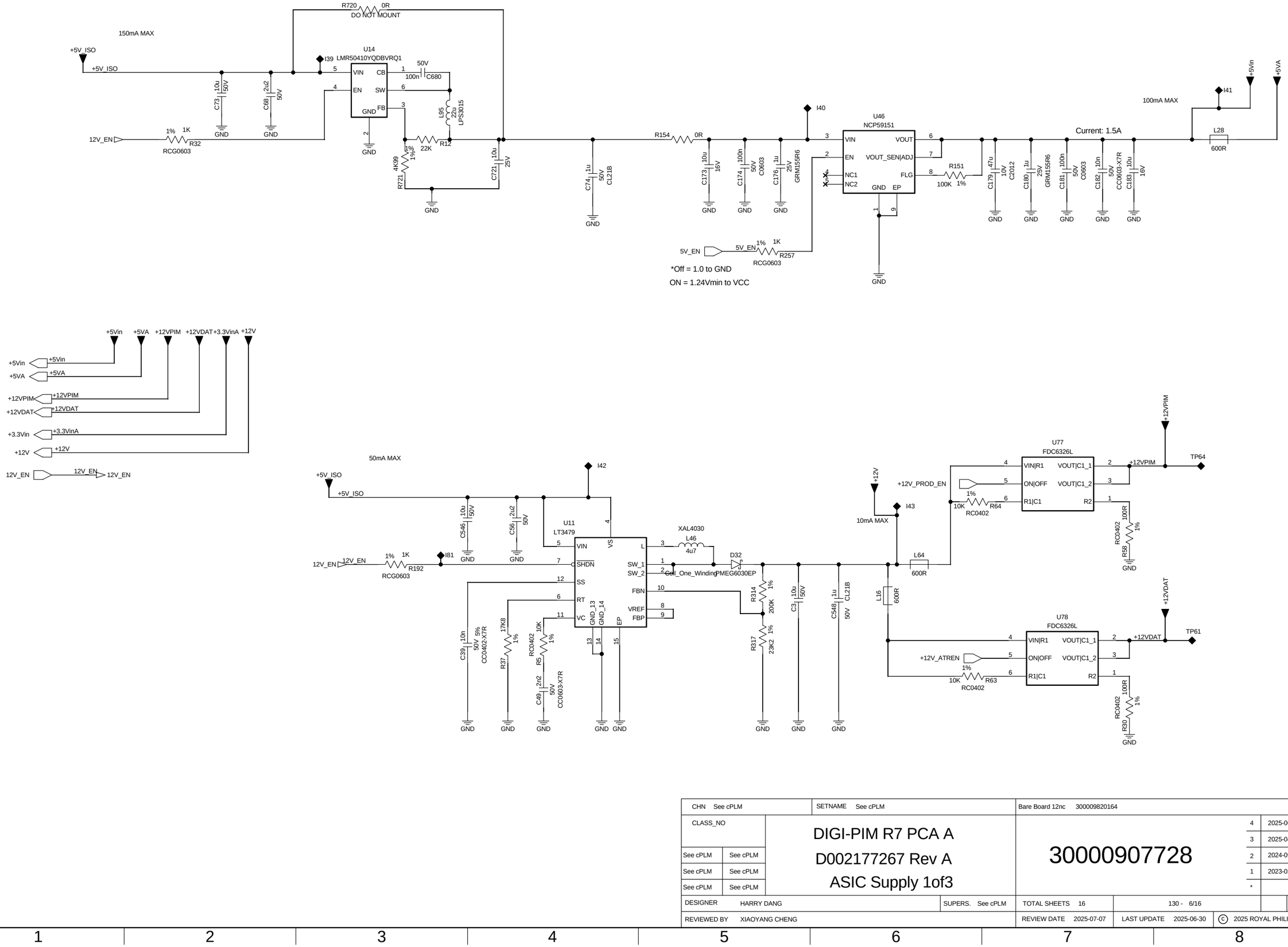
DGCTRIG	TGCA0	TGCA1	Time Constant
L	L	L	0.83usec
L	H	L	1.25usec
L	L	H	2.12usec
L	H	H	3.97usec

Notes: -tgcA0 and tgc A1 is from FPGA

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A RF_frontend_2of2	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 5/16 A3
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CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A ASIC Supply 1of3	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 6/16 A3
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PHILIPS

*10-75VDC HV output
Current Limit: 10mA
10K = 200uA
20K = 100uA
2.49K = 800uA
806 = 2.5mA
392 = 5mA

*Vr:100V, IF: 200mA

*392 Sense Resistor makes COMP 10K and 0.47uF

*806 Sense Resistor makes COMP 20K and 0.22uF

4.4msec

*Id 170mA

Sets adjust Span

Sets min HVout

*FB Traces to be as short as possible

*Low = Current Limit

*Window Comparator for 3.3VCATH
GAIN = x10
1.50V = 15mA Output
1.25V = 12mA Output
1.00V = 10mA Output
0.50V = 5mA Output
Ref Volt are 1.25V to 0V.

Current: 10mA

Current: 0.3A

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A ASIC Supply 2of3	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 7/16 A3
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Pre_HV	U12_pin4
30VDC	1.24V
31VDC	1.28V
35VDC	1.44V
38VDC	1.57V
40VDC	1.65V
45VDC	1.86V
48VDC	1.98V

Pre_HV	U13_pin3
48VDC	1.26V
47VDC	1.23V
42VDC	1.10V
40VDC	1.05V
38VDC	0.99V
35VDC	0.92V
30VDC	0.78V

+HV, R55 0R, I53, 1M 1%, R48, 43K2 0.1%, R43, C62 470n 10V C0402-X5R, C61 470n 10V C0402-X5R, GND

+3.3Vin, C54 10n 50V GRM155R7, C47 10u 25V GRM188R6, GND

U12 TLV3012 V+ V- REF, Vref: 1.242V, I54, PRDHV_UV_ACTL → PRDHV_UV_ACTL, Fault Under 30V

I58, RC0603 R84 47K 1%, I59, I3, C142 10n 50V GRM155R7, GND

1M 1%, R49, GND

U16 AD5310R: VOUT VREF VDD VLOGIC GND SDI SCLK SYNC RESET LDAC, +3.3Vin, C100 1u 25V GRM155R6, C103 10u 16V, GND

HV_UVFAULT_SYNC, HV_UVFAULT_SCLK, HV_UVFAULT_DIN → HV_UVFAULT_SYNC, HV_UVFAULT_SCLK, HV_UVFAULT_DIN

HV_OVFAULT_DIN → HV_OVFAULT_DIN, HV_OVFAULT_SCLK → HV_OVFAULT_SCLK, HV_OVFAULT_SYNC → HV_OVFAULT_SYNC

U26 AD5310R: SDI SCLK SYNC RESET LDAC VLOGIC VDD VOUT VREF GND, +3.3Vin, C125 1u 25V GRM155R6, C127 10u 16V, GND

RC0603 R85 47K 1%, I56, I4, C598 10n 50V GRM155R7, GND

I55

27K 1%, R44, C64 470n 10V C0402-X5R, C63 470n 10V C0402-X5R, GND

+3.3Vin, GRM155R7 C55 10n 50V, Digipot added for modular over/under voltage check, C46 10u 25V GRM188R6, GND

U13 TLV3012 V+ V- REF, Vref: 1.242V, Shutdown Over 48V, PRDHV_OV_ACTL → PRDHV_OV_ACTL, I57

+3.3Vin, C60 1u 25V GRM155R6, C58 10u 16V, GND

*Window Comparator for 3.3VCATH
GAIN = x10
1.50V = 15mA Output
1.25V = 12mA Output
1.00V = 10mA Output
0.50V = 5mA Output
Ref Volt are 1.25V to 0V.

+3.3Vin, C51 10n 50V CC0603-X7R, C45 10u 16V, GND

+5Vin, I85, C90 1u 25V GRM155R6, GND

*Power dissipation= 85mW @ 100mA
*50mV @ 100mA dropout Voltage
*Power Draw: 500mA Max

U23 TPS795: IN1 IN2 EN GND EP OUT1 OUT2 NR|FB NC

CATH3P3V_RAIL_EN, 1% 1K, RCG0603 R258

WF06Q 54K9 1% R46, C65 15p 50V, C77 1u 25V GRM155R6, C78 100n 50V C0603, C79 1u 25V GRM155R6, GND

TPS7950_OUT → I60

30K1 0.1% R52, GND

U19 LMP8646: IN+ IN- V+ V- VOUT RG

RC0603 10R 1% R53, Current: 0.5A, +3.3VCATH, C81 100n 50V C0603, C80 1u 25V GRM155R6, GND

RC0603 R38 1% 49R9, RC0603 49K9 1% R39, C66 100p 50V CC0402-NPO, RC0603 10K 1% R36, C59 1u 25V GRM155R6, GND

U15 TLV3012 V+ V- REF, I61, CATH3P3V_OI_ACTL

RC0603 R179 10K 1%, I62, 3V3MON, RC0603 10K 1% R180, GND

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A ASIC Supply 3of3	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 8/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30

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PHILIPS

*DOT FARADAY CAGE

3rd order Elliptic
Low Pass
Fc=43 MHz
Fstop=64.5 MHz
0.5dB ripple

Filter capacitors 0805 for higher Q-factor

*Clock to be placed close to FPGA.

*Sense Pin to VDD sets +/-1V input range

*REF Pins are ADC high/low reference

*Output Current Level Programmable Optional Internal 100ohm termination resistor between LVDS pairs

*Programmable Shutdown/sleep mode

105Msps; Sleep=0.5mW

*MAX SNR 73.4dB

*MAX Power=182mW

*Over/under flow pins when occured

* Add Multiple vias for OGND pin

*place close to ADC and repeater U69

Close to the FBGA

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A ADC RF	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 9/16 A3
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PHILIPS

FPGA_CTRL

P/N, N/P
SCK, SDAT, /DACS
DGCTRIG2, DGCTRIG
RXAMPSW
5V_EN
NOPROBE
EEPROM
CATH3P3V_RAIL_EN, LGCHV_OI_ACTL, 12V_EN, HVDACCSI, HVDACSCI, HVDACSDI, HVCLIF
HVADJ_DIN, HVADJ_SCLK, HVADJ_SYNC
HV_OVFAULT_DIN, HV_OVFAULT_SCLK, HV_OVFAULT_SYNC
HV_UVFAULT_DIN, HV_UVFAULT_SCLK, HV_UVFAULT_SYNC
LGCHV_RAIL_EN, XMIT
STAT_LO, STAT_HI
CATHSTR, CATH3P3V_OI_ACTL
PRDHV_RAIL_EN
TGCA0, TGCA1
TR_SWI_CONTROL
3V3MON
DIR_R/L
DAT+, DAT-
/CLOCK

PRDHV_OI_ACTL, PRDHV_CRBR_EN, XF3, PRDHV_UV_ACTL, PRDHV_OV_ACTL
XF7, XF8, XF9
CCLK, FRES
+1.2Vin, +3.3Vcco
+12V_PROD_EN, +12V_ATREN
EN_VMAG, SYNC_300
100MHz_SYS_CLK+, 100MHz_SYS_CLK-
CLK
FGA_PD3, FGA_PD1_N, FGA_PD2_N

SHT 11-12

FPGA_DATA

ADCP0, ADCN0, ADCP1, ADCN1, ADCP2, ADCN2, ADCP3, ADCN3, ADCP4, ADCN4, ADCP5, ADCN5, ADCP6, ADCN6
CLK+_ADC, CLK-_ADC, OF_+, OF_-, /CS_ADC, SCK_ADC, SDI_ADC, SDO_ADC
CCLK, FRES
+1.2Vin, +3.3Vcco
+12V_PROD_EN, +12V_ATREN
EN_VMAG, SYNC_300
100MHz_SYS_CLK+, 100MHz_SYS_CLK-

PHY_TX_CLK, PHY_TX_CTRL, PHY_TXD0, PHY_TXD1, PHY_TXD2, PHY_TXD3
PHY_RX_CLK, PHY_RX_CTRL, PHY_RXD0, PHY_RXD1, PHY_RXD2, PHY_RXD3
PHY_MDC, PHY_MDIO
PHY_RESETn
TestIN+, TestIN-
+1.0Vin, +1.8Vin, +2.5Vin

SHT 13-14

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A Digital Processing	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 10/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30

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PHILIPS

(3.3V)

U32
XC7A200T
BANK34

IO_0_34 T3 CFG_BIT5 → CFG_BIT5
IO_L1P_T0_34 T1 EN_VMAG → EN_VMAG
IO_L1N_T0_34 U1 FGA_PD1_N → FGA_PD1_N
IO_L2P_T0_34 U2 SDAT → SDAT
IO_L2N_T0_34 V2 SCK → SCK
IO_L3P_T0_DQS_34 R3 /DACS → /DACS
IO_L3N_T0_DQS_34 R2 DGCTRIG → DGCTRIG
IO_L4P_T0_34 W2 RXAMPSW → RXAMPSW
IO_L4N_T0_34 Y2 DGCTRIG2 → DGCTRIG2
IO_L5P_T0_34 W1 5V_EN → 5V_EN
IO_L5N_T0_34 Y1 NOPROBE → NOPROBE
IO_L6P_T0_34 U3 FGA_PD2_N → FGA_PD2_N
IO_L6N_T0_VREF_34 V3 12V_EN → 12V_EN
IO_L7P_T1_34 AA1 PRDHV_OI_ACTL → PRDHV_OI_ACTL
IO_L7N_T1_34 AB1 PRDHV_CRBR_EN → PRDHV_CRBR_EN
IO_L8P_T1_34 AB3
IO_L8N_T1_34 AB2 CATH3P3V_RAIL_EN → CATH3P3V_RAIL_EN
IO_L9P_T1_DQS_34 Y3 XF3 → XF3
IO_L9N_T1_DQS_34 AA3 PRDHV_UV_ACTL → PRDHV_UV_ACTL
IO_L10P_T1_34 AA5 LGCHV_OI_ACTL → LGCHV_OI_ACTL
IO_L10N_T1_34 AB5 FGA_PD3 → FGA_PD3
IO_L11P_T1_SRCC_34 Y4 P/N → P/N
IO_L11N_T1_SRCC_34 AA4 N/P → N/P
IO_L12P_T1_MRCC_34 V4 SYNC_300 → SYNC_300
IO_L12N_T1_MRCC_34 W4 TGCA0 → TGCA0
IO_L13P_T2_MRCC_34 R4 TGCA1 → TGCA1
IO_L13N_T2_MRCC_34 T4
IO_L14P_T2_SRCC_34 T5 XF9 → XF9
IO_L14N_T2_SRCC_34 U5 ALARM_EXT_STRAP
IO_L15P_T2_DQS_34 W6 → TR_SWI_CONTROL
IO_L15N_T2_DQS_34 W5 CFG_BIT9 → CFG_BIT9
IO_L16P_T2_34 U6 DAC_CS → HVDACSCI HV DAC Chip-Select Input
IO_L16N_T2_34 V5 → HVDACSCI HV DAC Serial Clock Input
IO_L17P_T2_34 R6 → HVDACSDI HV DAC Serial Data Input
IO_L17N_T2_34 T6 → HVCLIF HV Current Limit Indicator Flag
IO_L18P_T2_34 Y6 HVADJ_DIN → HVADJ_DIN
IO_L18N_T2_34 AA6 HVADJ_SCLK → HVADJ_SCLK
IO_L19P_T3_34 V7 LGCHV_RAIL_EN → LGCHV_RAIL_EN
IO_L19N_T3_VREF_34 W7 HVADJ_SYNC → HVADJ_SYNC
IO_L20P_T3_34 AB7 rx_host_bridge → rx_host_bridge
IO_L20N_T3_34 AB6 tx_host_bridge → tx_host_bridge
IO_L21P_T3_DQS_34 V9 HV_OVFAULT_DIN → HV_OVFAULT_DIN
IO_L21N_T3_DQS_34 V8 HV_OVFAULT_SCLK → HV_OVFAULT_SCLK
IO_L22P_T3_34 AA8 HV_OVFAULT_SYNC → HV_OVFAULT_SYNC
IO_L22N_T3_34 AB8 HV_UVFAULT_DIN → HV_UVFAULT_DIN
IO_L23P_T3_34 Y8 +12V_PROD_EN → +12V_PROD_EN
IO_L23N_T3_34 Y7 +12V_ATREN → +12V_ATREN
IO_L24P_T3_34 W9 HV_UVFAULT_SCLK → HV_UVFAULT_SCLK
IO_L24N_T3_34 Y9 HV_UVFAULT_SYNC → HV_UVFAULT_SYNC
IO_25_34 U7

+3.3V_PHY
U53 BAT54TW
R173 RC0603 100R 1% → EEPROM_TXRX
GND
CFG_BIT8 → CFG_BIT8
R353 1K DO NOT MOUNT
U47 2N7002DW
+10VCATH R145 RC0603 100K 1%
U45 2N7002DW
CLK → CLK
I90 → +3.3Vcco

U10
SN74LV1T34
+5Vin
I86
iXMIT → XMIT
C31 100n 16V CC0402-X7R
C42 1n 50V 04025C
C32 1u 25V GRM155R6

Q15 BSS84AKS
U47 2N7002DW
Q16 SI1926DL-T1-E3

RC0402 R189 100R DO NOT MOUNT 1%
+3.3Vin
+12VDAT
4K7 1% R187
RC0402 10K 1% R188
RC0402 10K 1% R184
BSS84AKS Q15
EEPROM_TXRX
2N7002DW Q17
Q17 2N7002DW
C239 470n 50V CC0402-X7R
/PGM → /PGM
Q16
R181 RC0402 100R 1% → EEPROM
EEPROM

+3.3Vin
R217 1% 10K RC0402
R215 100R 1% → FW_LED2
R216 100R 1% → FW_LED3
U48 2N7002DW
D5 BLUE/ORANGE
330R 1% R211
330R 1% R214
+5Vin

(3.3V)

U32
XC7A200T
BANK35

IO_0_35 F4 CFG_BIT4 → CFG_BIT4
IO_L1P_T0_AD4P_35 B1 +5Vin_MEAS → +5Vin_MEAS
IO_L1N_T0_AD4N_35 A1
IO_L2P_T0_AD12P_35 C2 +12V_MEAS → +12V_MEAS
IO_L2N_T0_AD12N_35 B2
IO_L3P_T0_DQS_AD5P_35 E1 +HV_MEAS → +HV_MEAS
IO_L3N_T0_DQS_AD5N_35 D1
IO_L4P_T0_35 E2 CathStarDET → CathStarDET
IO_L4N_T0_35 D2 CATH3P3V_OI_ACTL → CATH3P3V_OI_ACTL
IO_L5P_T0_AD13P_35 G1 9.0VDC_MEAS → 9.0VDC_MEAS
IO_L5N_T0_AD13N_35 F1
IO_L6P_T0_35 F3 PRDHV_OV_ACTL → PRDHV_OV_ACTL
IO_L6N_T0_VREF_35 E3 PRDHV_RAIL_EN → PRDHV_RAIL_EN
IO_L7P_T1_AD6P_35 K1 7.0VDC_MEAS → 7.0VDC_MEAS
IO_L7N_T1_AD6N_35 J1
IO_L8P_T1_AD14P_35 H2 +3.3VCATH_MEAS → +3.3VCATH_MEAS
IO_L8N_T1_AD14N_35 G2
IO_L9P_T1_DQS_AD7P_35 K2 3V3MON → 3V3MON
IO_L9N_T1_DQS_AD7N_35 J2
IO_L10P_T1_AD15P_35 J5 DIR_R/L → DIR_R/L
IO_L10N_T1_AD15N_35 H5
IO_L11P_T1_SRCC_35 H3 CFG_BIT10 → CFG_BIT10
IO_L11N_T1_SRCC_35 G3 CFG_BIT6 → CFG_BIT6
IO_L12P_T1_MRCC_35 H4 /PGM → /PGM
IO_L12N_T1_MRCC_35 G4 CFG_BIT11 → CFG_BIT11
IO_L13P_T2_MRCC_35 K4 CFG_BIT12 → CFG_BIT12
IO_L13N_T2_MRCC_35 J4 CFG_BIT13 → CFG_BIT13
IO_L14P_T2_SRCC_35 L3 CFG_BIT14 → CFG_BIT14
IO_L14N_T2_SRCC_35 K3 CFG_BIT15 → CFG_BIT15
IO_L15P_T2_DQS_35 M1 XF7 → XF7
IO_L15N_T2_DQS_35 L1 XF8 → XF8
IO_L16P_T2_35 M3 CFG_BIT7 → CFG_BIT7
IO_L16N_T2_35 M2 STAT_HI → STAT_HI
IO_L17P_T2_35 K6 STAT_LO → STAT_LO
IO_L17N_T2_35 J6 DAT+ → DAT+
IO_L18P_T2_35 L5 DAT- → DAT-
IO_L18N_T2_35 L4 /CLOCK → /CLOCK
IO_L19P_T3_35 N4 FW_LED0 → FW_LED0
IO_L19N_T3_VREF_35 N3 FW_LED1 → FW_LED1
IO_L20P_T3_35 R1 FW_LED2 → FW_LED2
IO_L20N_T3_35 P1 FW_LED3 → FW_LED3
IO_L21P_T3_DQS_35 P5 HW_LED4 → HW_LED4
IO_L21N_T3_DQS_35 P4 HW_LED5 → HW_LED5
IO_L22P_T3_35 P2 HW_LED6 → HW_LED6
IO_L22N_T3_35 N2 HW_LED7 → HW_LED7
IO_L23P_T3_35 M6 CFG_BIT0 → CFG_BIT0
IO_L23N_T3_35 M5 CFG_BIT1 → CFG_BIT1
IO_L24P_T3_35 P6 CFG_BIT2 → CFG_BIT2
IO_L24N_T3_35 N5 CFG_BIT3 → CFG_BIT3
IO_25_35 L6

+3.3Vin
C113 100n 16V CC0402-X7R
RC0402 4K99 1% R77 → CathStarDET
D14 BAV99
24K9 1% R69
WR04 1K 1% R67
C102 100n 16V CC0402-X7R
Q10 MMBT3904
CATHSTR → CATHSTR
High: 3.3VDC CathStar Connected
Low: 0VDC No Catheter Connected

(0.0V)

U32
XC7A200T
BANK16

IO_0_16 F15
IO_L1P_T0_16 F13
IO_L1N_T0_16 F14
IO_L2P_T0_16 F16
IO_L2N_T0_16 E17
IO_L3P_T0_DQS_16 C14
IO_L3N_T0_DQS_16 C15
IO_L4P_T0_16 E13
IO_L4N_T0_16 E14
IO_L5P_T0_16 E16
IO_L5N_T0_16 D16
IO_L6P_T0_16 D14
IO_L6N_T0_VREF_16 D15
IO_L7P_T1_16 B15
IO_L7N_T1_16 B16
IO_L8P_T1_16 C13
IO_L8N_T1_16 B13
IO_L9P_T1_DQS_16 A15
IO_L9N_T1_DQS_16 A16
IO_L10P_T1_16 A13
IO_L10N_T1_16 A14
IO_L11P_T1_SRCC_16 B17
IO_L11N_T1_SRCC_16 B18
IO_L12P_T1_MRCC_16 D17
IO_L12N_T1_MRCC_16 C17
IO_L13P_T2_MRCC_16 C18
IO_L13N_T2_MRCC_16 C19
IO_L14P_T2_SRCC_16 E19
IO_L14N_T2_SRCC_16 D19
IO_L15P_T2_DQS_16 F18
IO_L15N_T2_DQS_16 E18
IO_L16P_T2_16 B20
IO_L16N_T2_16 A20
IO_L17P_T2_16 A18
IO_L17N_T2_16 A19
IO_L18P_T2_16 F19
IO_L18N_T2_16 F20
IO_L19P_T3_16 D20
IO_L19N_T3_VREF_16 C20
IO_L20P_T3_16 C22
IO_L20N_T3_16 B22
IO_L21P_T3_DQS_16 B21
IO_L21N_T3_DQS_16 A21
IO_L22P_T3_16 E22
IO_L22N_T3_16 D22
IO_L23P_T3_16 E21
IO_L23N_T3_16 D21
IO_L24P_T3_16 G21
IO_L24N_T3_16 G22
IO_25_16 F21
GND

U32
XC7A200T
BANK_216

MGTPTXP3_216 D7
MGTPRXP3_216 D9
MGTPTXN3_216 C7
MGTPRXN3_216 C9
MGTPTXP2_216 B6
MGTPRXP2_216 B10
MGTPTXN2_216 A6
MGTPRXN2_216 A10
MGTREFCLK0N_216 E6
MGTREFCLK0P_216 F6
MGTRREF_216 F8
MGTREFCLK1P_216 E10
MGTREFCLK1N_216 D5
MGTPTXP1_216 D11
MGTPRXP1_216 C5
MGTPTXN1_216 C11
MGTPRXN1_216 B4
MGTPTXP0_216 B8
MGTPRXP0_216 A4
MGTPTXN0_216 A8
MGTPRXN0_216
GND

U32
XC7A200T
GND

W18 GND_W18 GND_G10 G10
V21 GND_V21 GND_G9 G9
V11 GND_V11 GND_G14 G14
V1 GND_V1 GND_G12 G12
U14 GND_U14 GND_G5 G5
Y15 GND_Y15 GND_G8 G8
Y5 GND_Y5 GND_G7 G7
W8 GND_W8 GND_G6 G6
T17 GND_T17 GND_F5 F5
U4 GND_U4 GND_E20 E20
T11 GND_T11 GND_F17 F17
T7 GND_T7 GND_F11 F11
T9 GND_T9 GND_E11 E11
R10 GND_R10 GND_E7 E7
R20 GND_R20 GND_E9 E9
R12 GND_R12 GND_D13 D13
P9 GND_P9 GND_E5 E5
P7 GND_P7 GND_E4 E4
R8 GND_R8 GND_D8 D8
P13 GND_P13 GND_D4 D4
P11 GND_P11 GND_D12 D12
N8 GND_N8 GND_C12 C12
N6 GND_N6 GND_C10 C10
P3 GND_P3 GND_C6 C6
N16 GND_N16 GND_D3 D3
L22 GND_L22 GND_C16 C16
L8 GND_L8 GND_B3 B3
L2 GND_L2 GND_C3 C3
K15 GND_K15 GND_B19 B19
M11 GND_M11 GND_B12 B12
M7 GND_M7 GND_AA22 AA22
K5 GND_K5 GND_AA12 AA12
M19 GND_M19 GND_AB19 AB19
K11 GND_K11 GND_AB9 AB9
K7 GND_K7 GND_A22 A22
J8 GND_J8 GND_AA2 AA2
H21 GND_H21 GND_A12 A12
H11 GND_H11 GND_A11 A11
H9 GND_H9 GND_A9 A9
H7 GND_H7 GND_A5 A5
J18 GND_J18 GND_A7 A7
J12 GND_J12 GND_A2 A2
J10 GND_J10 GND_A3 A3
H1 GND_H1
GND GND

+5Vin
9K1 R124
1K R115
C163 1n 25V C0402
+5Vin_MEAS → +5Vin_MEAS

+12V
15K R125
1K R116
C166 1n 25V C0402
+12V_MEAS → +12V_MEAS

+HV
590K R130
10K 1% RC0402 R138
C167 1n 25V C0402
+HV_MEAS → +HV_MEAS

9.0VDC
15K R126
1K R118
C168 1n 25V C0402
9.0VDC_MEAS → 9.0VDC_MEAS

7.0VDC
15K R127
1K R120
C169 1n 25V C0402
7.0VDC_MEAS → 7.0VDC_MEAS

+3.3VCATH
15K R129
1K 1% R122
C170 1n 25V C0402
+3.3VCATH_MEAS → +3.3VCATH_MEAS

J3
CFG_BIT8 2 1 CFG_BIT0 CFG_BIT0
CFG_BIT9 4 3 CFG_BIT1 CFG_BIT1
CFG_BIT10 6 5 CFG_BIT2 CFG_BIT2
CFG_BIT11 8 7 CFG_BIT3 CFG_BIT3
CFG_BIT12 10 9 CFG_BIT4 CFG_BIT4
CFG_BIT13 12 11 CFG_BIT5 CFG_BIT5
CFG_BIT14 14 13 CFG_BIT6 CFG_BIT6
CFG_BIT15 16 15 CFG_BIT7 CFG_BIT7

*Place Signals on the outside board GND on inside

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A FPGA_Control_1of2	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 11/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 © 2025 ROYAL PHILIPS N.V.

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PHILIPS

1 2 3 4 5 6 7 8

A B C D E F

+3.3Vcco
5%
RN1 2K2
D3 598-8070-107F GREEN
D4 598-8070-107F GREEN
FW_LED0
FW_LED1

+3.3Vcco
5%
RN2 2K2
D7 598-8070-107F GREEN
D8 598-8070-107F GREEN
D9 598-8070-107F GREEN
D10 598-8070-107F GREEN
HW_LED4
HW_LED5
HW_LED6
HW_LED7

+1.8Vin
J2 57202
JTAG_TMS
JTAG_TCK
JTAG_TDO
JTAG_TDI
GND

JTAG
+1.8Vin
L19 600R
avdd
C321 100n 16V CC0402-X7R
C320 470n 25V
+1.2Vin
C299 100n 16V CC0402-X7R
C159 10u 50V
GND

DONE
+1.8Vin
WR04 1K 1%
R141
D17 598-8070-107F GREEN
GND

+10VCATH
R377 15K
R378 1K
C599 1n 25V C0402
+10VCATH_MEAS
GND

U32
XC7A200T
BANK_0
DXN_0 N9
VCCADC_0 K10
GNDADC_0 K9
DXP_0 N10
VREFN_0 L9
VREFP_0 M10
VP_0 L10 +10VCATH_MEAS
VN_0 M9 GND
VCCBATT_0 E12
CCLK_0 L12 CCLK
TCK_0 V12 JTAG_TCK
TMS_0 T13 JTAG_TMS
TDO_0 U13 JTAG_TDO
TDI_0 R13 JTAG_TDI
INIT_B_0 U12 init_B
PROGRAM_B_0 N12
CFGBVS_0 U8
DONE_0 G11 FPGA_STATUS
M2_0 U9
M0_0 U11 +1.8Vin
M1_0 U10
GND
RC0805 R376 0R
GND

+1.8Vin
4K7 1% R209
+1.8Vin
4K7 1% R206
+1.8Vin
WR04 1K 1%
R137
init_B/ CRC-error
D16 598-8010-107F RED
FRES

+3.3V_PHY
4K7 1% R24
+3.3V_PHY
C227 1u 25V GRM155R6
C228 100n 16V CC0402-X7R
+3.3V_PHY
C22 1u 25V GRM155R6
C229 100n 16V CC0402-X7R
GND

U8
CP2102N
VBUS 8
VREGIN 7
RI|CLK 1
RST 9
TXD 21 tx_host_bridge
RXD 20 rx_host_bridge
DCD 24
DTR 23
DSR 22
RTS 19
CTS 18
D+ 3
D- 4
VDD 6
VIO 5
SUSPEND 17
SUSPEND 15
GPIO.0|TXT 14 TPAD4
GPIO.1|RXT 13 TPAD3
GPIO.2|RS485 12 TPAD2
GPIO.3|WAKEUP 11 TPAD1
NC_10 10
NC_16 16
GND EP 2 25
GND

J1
500075
GND GND
8.5V
D21 SP0503BAHTG
GND

+1.8Vin
C334 47u 10V GRM32-R7
GND
U32
XC7A200T
VCCO
VCCO_0_T12 T12
VCCO_0_F12 F12
VCCO_13_Y10 Y10
VCCO_13_W13 W13
VCCO_13_V16 V16
VCCO_13_AB14 AB14
VCCO_13_AA17 AA17
VCCO_14_Y20 Y20
VCCO_14_U19 U19
VCCO_14_T22 T22
VCCO_14_R15 R15
VCCO_14_P18 P18
VCCO_14_M14 M14
VCCO_15_N21 N21
VCCO_15_L17 L17
VCCO_15_K20 K20
VCCO_15_J13 J13
VCCO_15_H16 H16
VCCO_15_G19 G19
VCCO_16_F22 F22
VCCO_16_E15 E15
VCCO_16_D18 D18
VCCO_16_C21 C21
VCCO_16_B14 B14
VCCO_16_A17 A17
VCCO_34_W3 W3
VCCO_34_V6 V6
VCCO_34_T2 T2
VCCO_34_R5 R5
VCCO_34_AB4 AB4
VCCO_34_AA7 AA7
VCCO_35_N1 N1
VCCO_35_M4 M4
VCCO_35_J3 J3
VCCO_35_H6 H6
VCCO_35_F2 F2
VCCO_35_C1 C1
GND

+1.8Vin
C398 47u 10V GRM32-R7
C323 4u7 25V GRM188R6
C399 4u7 25V GRM188R6
GND
C400 470n 10V CC0402-X5R
C356 470n 10V CC0402-X5R
C420 470n 10V CC0402-X5R
C437 470n 10V CC0402-X5R
GND

+1.8Vin
C444 47u 10V GRM32-R7
C401 4u7 10V C1005
C324 4u7 10V C1005
GND
C402 470n 10V CC0402-X5R
C349 470n 10V CC0402-X5R
C325 470n 10V CC0402-X5R
C357 470n 10V CC0402-X5R
GND

+2.5Vin
C433 47u 10V GRM32-R7
C326 4u7 10V C1005
C346 4u7 10V C1005
GND
C403 470n 10V CC0402-X5R
C372 470n 10V CC0402-X5R
C384 470n 10V CC0402-X5R
C345 470n 10V CC0402-X5R
GND

+3.3Vcco
C257 47u 10V GRM32-R7
C271 4u7 10V C1005
C281 4u7 10V C1005
GND
C258 470n 10V CC0402-X5R
C270 470n 10V CC0402-X5R
C280 470n 10V CC0402-X5R
C260 470n 10V CC0402-X5R
GND

+3.3Vcco
C259 47u 10V GRM32-R7
C276 4u7 10V C1005
C254 4u7 10V C1005
GND
C272 470n 10V CC0402-X5R
C261 470n 10V CC0402-X5R
C277 470n 10V CC0402-X5R
C253 470n 10V CC0402-X5R
GND

+3.3Vcco +1.2Vin
+1.2Vin
+3.3Vcco

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A FPGA_Control_2of2	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 12/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30

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PHILIPS

(1.8V)

U32
XC7A200T
BANK_14

IO_0_14 P20
IO_L1P_T0_D00_MOSI_14 P22 → fd0
IO_L1N_T0_D01_DIN_14 R22 → fd1
IO_L2P_T0_D02_14 P21 → fd2
IO_L2N_T0_D03_14 R21 → fd3
IO_L3P_T0_DQS_PUDC_B_14 U22
IO_L3N_T0_DQS_EMCCLK_14 V22 → GND
IO_L4P_T0_D04_14 T21
IO_L4N_T0_D05_14 U21
IO_L5P_T0_D06_14 P19
IO_L5N_T0_D07_14 R19
IO_L6P_T0_FCS_B_14 T19 → fcs
IO_L6N_T0_D08_VREF_14 T20
IO_L7P_T1_D09_14 W21
IO_L7N_T1_D10_14 W22
IO_L8P_T1_D11_14 AA20
IO_L8N_T1_D12_14 AA21
IO_L9P_T1_DQS_14 Y21
IO_L9N_T1_DQS_D13_14 Y22
IO_L10P_T1_D14_14 AB21
IO_L10N_T1_D15_14 AB22
IO_L11P_T1_SRCC_14 U20
IO_L11N_T1_SRCC_14 V20
IO_L12P_T1_MRCC_14 W19 → 100MHz_SYS_CLK+
IO_L12N_T1_MRCC_14 W20 → 100MHz_SYS_CLK-
IO_L13P_T2_MRCC_14 Y18
IO_L13N_T2_MRCC_14 Y19
IO_L14P_T2_SRCC_14 V18
IO_L14N_T2_SRCC_14 V19
IO_L15P_T2_DQS_RDWR_B_14 AA19
IO_L15N_T2_DQS_DOUT_CSO_B_14 AB20
IO_L16P_T2_CSI_B_14 V17
IO_L16N_T2_A15_D31_14 W17
IO_L17P_T2_A14_D30_14 AA18
IO_L17N_T2_A13_D29_14 AB18
IO_L18P_T2_A12_D28_14 U17
IO_L18N_T2_A11_D27_14 U18
IO_L19P_T3_A10_D26_14 P14
IO_L19N_T3_A09_D25_VREF_14 R14
IO_L20P_T3_A08_D24_14 R18
IO_L20N_T3_A07_D23_14 T18
IO_L21P_T3_DQS_14 N17
IO_L21N_T3_DQS_A06_D22_14 P17
IO_L22P_T3_A05_D21_14 P15
IO_L22N_T3_A04_D20_14 R16
IO_L23P_T3_A03_D19_14 N13
IO_L23N_T3_A02_D18_14 N14
IO_L24P_T3_A01_D17_14 P16
IO_L24N_T3_A00_D16_14 R17
IO_25_14 N15

(2.5V)

U32
XC7A200T
BANK15

IO_0_15 J16
IO_L1P_T0_AD0P_15 H13 ADCP0 → ADCP0
IO_L1N_T0_AD0N_15 G13 ADCN0 → ADCN0
IO_L2P_T0_AD8P_15 G15 ADCP1 → ADCP1
IO_L2N_T0_AD8N_15 G16 ADCN1 → ADCN1
IO_L3P_T0_DQS_AD1P_15 J14 ADCP2 → ADCP2
IO_L3N_T0_DQS_AD1N_15 H14 ADCN2 → ADCN2
IO_L4P_T0_15 G17 ADCP3 → ADCP3
IO_L4N_T0_15 G18 ADCN3 → ADCN3
IO_L5P_T0_AD9P_15 J15 ADCP4 → ADCP4
IO_L5N_T0_AD9N_15 H15 ADCN4 → ADCN4
IO_L6P_T0_15 H17 ADCP5 → ADCP5
IO_L6N_T0_VREF_15 H18 ADCN5 → ADCN5
IO_L7P_T1_AD2P_15 J22 ADCP6 → ADCP6
IO_L7N_T1_AD2N_15 H22 ADCN6 → ADCN6
IO_L8P_T1_AD10P_15 H20
IO_L8N_T1_AD10N_15 G20 → GND
IO_L9P_T1_DQS_AD3P_15 K21 OF_+ → OF_+
IO_L9N_T1_DQS_AD3N_15 K22 OF_- → OF_-
IO_L10P_T1_AD11P_15 M21 → bit3
IO_L10N_T1_AD11N_15 L21
IO_L11P_T1_SRCC_15 J20 → bit2
IO_L11N_T1_SRCC_15 J21
IO_L12P_T1_MRCC_15 J19 CLK+_ADC → CLK+_ADC
IO_L12N_T1_MRCC_15 H19 CLK-_ADC → CLK-_ADC
IO_L13P_T2_MRCC_15 K18 → bit1
IO_L13N_T2_MRCC_15 K19 LED_R
IO_L14P_T2_SRCC_15 L19 LED_G → bit0
IO_L14N_T2_SRCC_15 L20 LED_B
IO_L15P_T2_DQS_15 N22
IO_L15N_T2_DQS_ADV_B_15 M22
IO_L16P_T2_A28_15 M18
IO_L16N_T2_A27_15 L18
IO_L17P_T2_A26_15 N18
IO_L17N_T2_A25_15 N19
IO_L18P_T2_A24_15 N20
IO_L18N_T2_A23_15 M20
IO_L19P_T3_A22_15 K13
IO_L19N_T3_A21_VREF_15 K14
IO_L20P_T3_A20_15 M13 SCL_ADC0 → SCL_ADC0
IO_L20N_T3_A19_15 L13 SDA_ADC0 → SDA_ADC0
IO_L21P_T3_DQS_15 K17 SCL_ADC1 → SCL_ADC1
IO_L21N_T3_DQS_A18_15 J17 SDA_ADC1 → SDA_ADC1
IO_L22P_T3_A17_15 L14
IO_L22N_T3_A16_15 L15
IO_L23P_T3_FOE_B_15 L16
IO_L23N_T3_FWE_B_15 K16
IO_L24P_T3_RS1_15 M15 TestIN+ → TestIN+
IO_L24N_T3_RS0_15 M16 TestIN- → TestIN-
IO_25_15 M17

RC0402 10K 1% R324
RC0402 10K 1% R325

U74
SHTC3
VDD SCL SDA VSS EP
C573 100n 16V CC0402-X7R

TEMP

+2.5Vin
Rev. 7
DO NOT MOUNT 4K7 1% RP20 RP21 RP22 RP23 RP24 RP25 RP26 RP27

+1.0Vin
C302 100n 16V CC0402-X7R
C464 100n 16V CC0402-X7R
C458 680u 6.3V T530
GND

C301 C490 C456 C296 C304 C453 C451 C453 C289 C287 C294 C319 4u7 10V C1005

C303 C300 C284 C461 C291 C457 C497 C292 C305 C293 C452 C499 C288 C295 470n 10V CC0402-X5R

+1.8Vin
C282 47u 10V GRM32-R7
C285 C431 C382 C427 4u7 10V C1005

C286 C443 C279 C432 C383 C440 C426 470n 10V CC0402-X5R

+5VA
D33 LTST-C19HE1WT RED-GREEN-BLUE
C600 1u 25V CC0603-X7R

C298 C283 C297 470n 10V CC0402-X5R
C469 100u 6.3V

+2.5Vin +1.8Vin +1.0Vin
+1.0Vin +1.8Vin +2.5Vin

U32
XC7A200T
VCC_INT_AUX_BRAM

VCCINT_T8 T8
VCCINT_T10 T10
VCCINT_R9 R9
VCCINT_R7 R7
VCCINT_P10 P10
VCCINT_P8 P8
VCCINT_N7 N7
VCCINT_M8 M8
VCCINT_K8 K8
VCCINT_L7 L7
VCCINT_J9 J9
VCCINT_J7 J7
VCCINT_H8 H8
VCCINT_H10 H10
VCCAUX_R11 R11
VCCAUX_P12 P12
VCCAUX_M12 M12
VCCAUX_K12 K12
VCCAUX_H12 H12
VCCBRAM_N11 N11
VCCBRAM_L11 L11
VCCBRAM_J11 J11

(1.8V)

U32
XC7A200T
BANK13

IO_0_13 Y17 PHY_RXD0 → PHY_RXD0
IO_L1P_T0_13 Y16 PHY_RXD1 → PHY_RXD1
IO_L1N_T0_13 AA16 PHY_RXD2 → PHY_RXD2
IO_L2P_T0_13 AB16 PHY_RXD3 → PHY_RXD3
IO_L2N_T0_13 AB17 PHY_RX_CTRL → PHY_RX_CTRL
IO_L3P_T0_DQS_13 AA13 R369 22R 1% PHY_TXD0 → PHY_TXD0
IO_L3N_T0_DQS_13 AB13 R370 22R 1% PHY_TXD1 → PHY_TXD1
IO_L4P_T0_13 AA15 R371 22R 1% PHY_TXD2 → PHY_TXD2
IO_L4N_T0_13 AB15 R372 22R 1% PHY_TXD3 → PHY_TXD3
IO_L5P_T0_13 Y13 R368 22R 1% PHY_TX_CTRL → PHY_TX_CTRL
IO_L5N_T0_13 AA14 PHY_MDIO → PHY_MDIO
IO_L6P_T0_13 W14 PHY_MDC → PHY_MDC
IO_L6N_T0_VREF_13 Y14 PHY_RESETn → PHY_RESETn
IO_L7P_T1_13 AB11
IO_L7N_T1_13 AB12 R367 22R 1% PHY_TX_CLK → PHY_TX_CLK
IO_L8P_T1_13 AA9
IO_L8N_T1_13 AB10
IO_L9P_T1_DQS_13 AA10 /CS_ADC → /CS_ADC
IO_L9N_T1_DQS_13 AA11 SCK_ADC → SCK_ADC
IO_L10P_T1_13 V10 SDI_ADC → SDI_ADC
IO_L10N_T1_13 W10 SDO_ADC → SDO_ADC
IO_L11P_T1_SRCC_13 Y11
IO_L11N_T1_SRCC_13 Y12
IO_L12P_T1_MRCC_13 W11 PHY_RX_CLK → PHY_RX_CLK
IO_L12N_T1_MRCC_13 W12
IO_L13P_T2_MRCC_13 V13
IO_L13N_T2_MRCC_13 V14
IO_L14P_T2_SRCC_13 U15
IO_L14N_T2_SRCC_13 V15
IO_L15P_T2_DQS_13 T14
IO_L15N_T2_DQS_13 T15
IO_L16P_T2_13 W15
IO_L16N_T2_13 W16
IO_L17P_T2_13 T16
IO_L17N_T2_13 U16

U80 U81 U82
BCV61 T1 T2
R379 R380 R381 180R
LED_R LED_G LED_B

U42
MT25QU256ABA
CCLK B2 CLK
fcs C2 CS
fd2 C4 WP|DQ2
FRES A4 RESET
fd3 D4 HOLD|DQ3
A2 NC
A3 RFU_A3
A5 RFU_A5
B1 RFU_B1
B5 RFU_B5
C1 RFU_C1
VCC B4
DQ0 D3 → fd0
DQ1 D2 → fd1
RFU_C3 C3
RFU_C5 C5
RFU_D1 D1
RFU_D5 D5
RFU_E1 E1
RFU_E2 E2
RFU_E3 E3
RFU_E4 E4
RFU_E5 E5
VSS B3

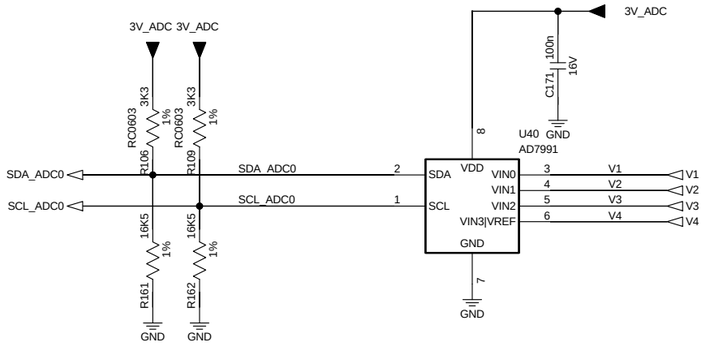
R233 2K4 1%
R333 4K7 1%
R334 4K7 1%
C160 100n 16V CC0402-X7R

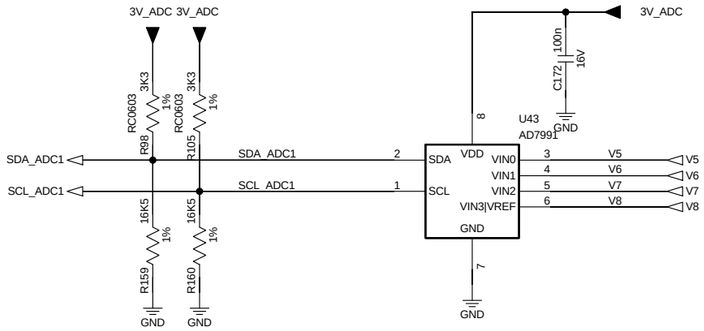
U32
XC7A200T
MGTA_VTT_VCC

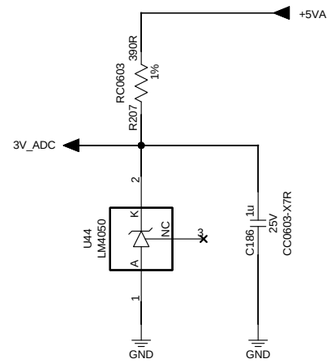
MGTAVCC_F7 F7
MGTAVCC_E8 E8
MGTAVCC_F9 F9
MGTAVCC_D10 D10
MGTAVCC_D6 D6
MGTAVTT_C4 C4
MGTAVTT_B11 B11
MGTAVTT_B9 B9
MGTAVTT_C8 C8
MGTAVTT_B7 B7
MGTAVTT_B5 B5

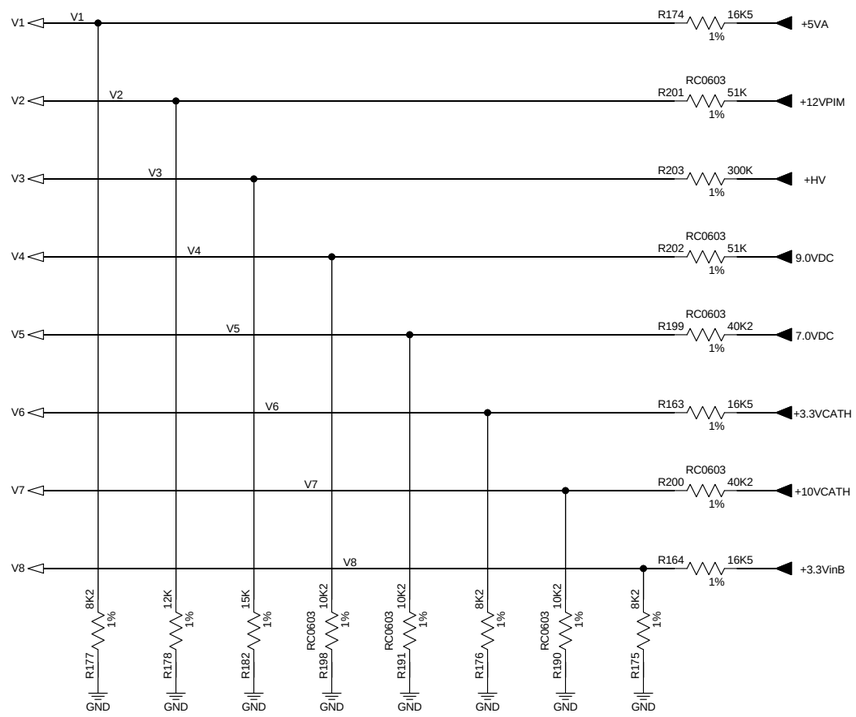
CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A FPGA DATA	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM See cPLM			
See cPLM See cPLM			
See cPLM See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 13/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 © 2025 ROYAL PHILIPS N.V.

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CHN See cPLM		SETNAME See cPLM		Bare Board 12nc 300009820164	
CLASS_NO		DIGI-PIM R7 PCA A D002177267		30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 * * .
See cPLM	See cPLM				
See cPLM	See cPLM				
See cPLM	See cPLM				
DESIGNER HARRY DANG		SUPERS. See cPLM	TOTAL SHEETS 16	130 - 14/16	A3
REVIEWED BY XIAOYANG CHENG			REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30	© 2025 ROYAL PHILIPS N.V.

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PHILIPS

*ESD/EFT Suppression

U60 — SP3304NUTG — 11.5V 300W

C380 100n 16V CC0402-X7R

+TD1_A / -TD1_A / +TD2_B / -TD2_B / +TD3_C / -TD3_C / +TD4_D / -TD4_D

U59 — SP3304NUTG — 11.5V

C353 100n 16V CC0402-X7R

*ESD/EFT Suppression

FB2 600R WE-CBF

+1.8V_PHY / +1.8V_FILT — Current: 140mA — I92

+3.3V_PHY — PHY_DVDD — I93

U36 88E1518

AVDD18_1 / AVDD18_2 / AVDD18_OUT / AVDDC18 / AVDD33_1 / AVDD33_2 / DVDD_1 / DVDD_2 / DVDD_OUT / VDDO_1 / VDDO_2 / VDDO_3

+TD1_A	Orange-White_ABC	24	MDIP0
-TD1_A	Orange_ABC	23	MDIN0
+TD2_B	Green-White_ABC	20	MDIP1
-TD2_B	Blue_ABC	19	MDIN1
+TD3_C	Blue-White_ABC	18	MDIP2
-TD3_C	Green_ABC	17	MDIN2
+TD4_D	Brown-White_ABC	14	MDIP3
-TD4_D	Brown_ABC	13	MDIN3
PHY_RSET	R229 4K99 1%	25	RSET
PHY_MDC	R208 4K7 1%	4	MDC
		5	MDIO
PHY_LED0		10	LED0
PHY_LED1		9	LED1
PHY_LED2		8	LED2/INT
	I124	6	CLK125
	DO NOT MOUNT L66 600R	31	REG_IN
	C165 220n 16V GRM155R7	32	REGCAP1
		35	REGCAP2
		49	EP

TX_CLK	47	PHY_TX_CLK	PHY_TX_CLK
TX_CTRL	2	PHY_TX_CTRL	PHY_TX_CTRL
TXD0	44	PHY_TXD0	PHY_TXD0
TXD1	45	PHY_TXD1	PHY_TXD1
TXD2	48	PHY_TXD2	PHY_TXD2
TXD3	1	PHY_TXD3	PHY_TXD3
RX_CLK	40	R361 22R 1%	PHY_RX_CLK
RX_CTRL	37	R362 22R 1%	PHY_RX_CTRL
RXD0	38	R363 22R 1%	PHY_RXD0
RXD1	39	R364 22R 1%	PHY_RXD1
RXD2	41	R365 22R 1%	PHY_RXD2
RXD3	42	R366 22R 1%	PHY_RXD3
HSDACP	27		
HSDACN	26		
XTAL_IN	29		
XTAL_OUT	28		
RESET	12		
CONFIG	11		

PHY_MDIO

+3.3V_PHY — I124 — DO NOT MOUNT L66 600R

PHY_DVDD / +1.8V_PHY / +3.3V_PHY

RC0402 R222 100R 1% DO NOT MOUNT

R224 49R9 1% DO NOT MOUNT — R223 49R9 1% DO NOT MOUNT

Y1 NX2016SA 25MHz

C154 8p2 50V CC0805-NPO — C155 8p2 50V CC0805-NPO

+1.8V_PHY — R323 4K7 1% — PHY_RESETn

+1.8V_PHY — R86 0R — R78 1K 1% DO NOT MOUNT

Current: 1.7A

5V_SYSIN — C190 10u 50V C1210-X7R — C191 10u 50V C1210-X7R — C192 4u7 25V GRM188R6 — C193 1u 25V GRM155R6 — C194 100n 50V C0603 — RC0805 R128 0R — C580 1u 25V GRM155R6 — C579 10n 50V GRM155R7 — L15 600R MPZ2012 — +5V_ISO

+3.3V_PHY — RC0603 R197 1K 1% — D13 598-8010-107F RED — Q8 NDS331N — PHY_LED0 — RC0603 R196 100K 1%

+3.3V_PHY — RC0603 R193 1K 1% — D11 598-8010-107F RED — Q6 NDS331N — PHY_LED1 — RC0603 R70 100K 1%

+3.3V_PHY — I91 — RC0603 R195 1K 1% — D12 598-8010-107F RED — Q7 NDS331N — PHY_LED2 — RC0603 R194 100K 1%

PHY_DVDD — *PHY_DVDD Supplied Internally — C419 4u7 25V GRM188R6 — C332 C331 C395 C396 C147 100n 16V CC0402-X7R

+1.8V_PHY — C381 4u7 25V GRM188R6 — C120 C333 C355 C397 C425 100n 16V CC0402-X7R

+1.8V_FILT — C318 4u7 25V GRM188R6 — C348 C371 100n 16V CC0402-X7R — C123 1n 16V CC0402-X7R

+3.3V_PHY — C164 4u7 25V GRM188R6 — C99 C153 100n 16V CC0402-X7R

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Rev A	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 *
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 15/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30

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PHILIPS

PMIC output Power up sequence: 1V0_EN, 1V2_EN, 1V8_EN, 3V3_EN, 2V5_EN

U49 ADP5052

U35 ADM6710

U73 CSD13385F5

U70 CSD13385F5

text

CHN See cPLM	SETNAME See cPLM	Bare Board 12nc 300009820164	
CLASS_NO	DIGI-PIM R7 PCA A D002177267 Local Power	30000907728	4 2025-06-30 3 2025-04-07 2 2024-09-05 1 2023-02-10 * .
See cPLM / See cPLM			
See cPLM / See cPLM			
See cPLM / See cPLM			
DESIGNER HARRY DANG	SUPERS. See cPLM	TOTAL SHEETS 16	130 - 16/16 A3
REVIEWED BY XIAOYANG CHENG		REVIEW DATE 2025-07-07	LAST UPDATE 2025-06-30 © 2025 ROYAL PHILIPS N.V.